Discrete Planning

Vikas Dhiman

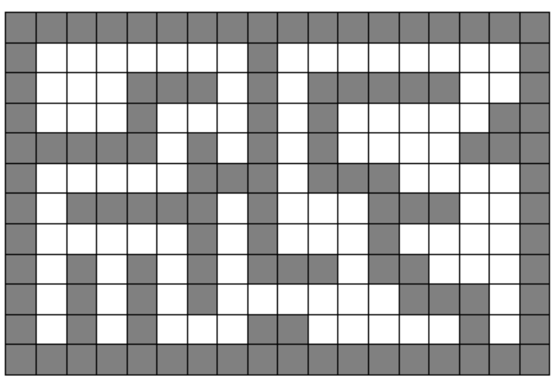
Friday 19th September, 2025

Open in COLAB

1 Required readings

- Ch 7 and 23, Carmen's Intro to Algorithms
- Breadth-first search and its uses (article) | Khan Academy
- PythonRobotics/dijkstra.py
- Discrete Planning by Lavalle
- 3.5.2 of Russel and Norvig: Artificial Intelligence

2 Discrete planning in Robotics



2.0.1 Abstraction of a planning problem

1. State space $\mathbf{s}_t \in \mathcal{S}$. Property of a robot that changes with time and represents all the information needed for future planning (Markovian property).

For example, 2D coordinate of a grid $\mathbf{s} = (x, y)$. If orientation of a robot matters, for example, a car that can only move forward or backwards and needs to turn before it can move left or right, a state $\mathbf{s} = (x, y, \theta)$. What would be state for a drone or a plane that can fly in 3D and rotate in 3D?

2. Action space per state $\mathbf{u} \in \mathcal{U}(\mathbf{s})$. The choices robot can make. For example, up, down, left right movement can be encoded as $\mathcal{U}(\mathbf{s}_t) = \{(0, -1), (0, 1), (1, 0), (-1, 0)\}$.

3. State transition function $\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t)$. For example, the up-down-left-right action can be combined as addition to get the next state $\mathbf{s}_{t+1} = \mathbf{s}_t + \mathbf{u}_t$. When the system is stochastic, then it also known as state transition probabilities $P(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{u}_t)$. In control systems, also known as system dynamics.

4. Initial State $\mathbf{s}_I \in \mathcal{S}$

5. Goal states $\mathbf{s}_G \subseteq \mathcal{S}$

2.0.2 A Graph

A graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$ is defined by a set of vertices $\mathcal{V}$ and a set of edges $\mathcal{E}$ such that each edge $e \in \mathcal{E}$ is formed by a pair of start and end vertices $e = (v_s, v_e), v_s \in \mathcal{V}, v_e \in \mathcal{V}$. The first vertex is called the start of the edge $v_s = \text{start}(e)$ and second vertex is called the end $v_e = \text{end}(e)$.

A discrete planning problem can be converted into a graph by definiting

1. Vertices as the state space $\mathcal{V} = \mathcal{S}$.

2. The action space at each state as the edges connected to that vertex/state, $\mathcal{U}(\mathbf{s}_t) = \{(\mathbf{s}_t, \mathbf{s}_j) \mid (\mathbf{s}_t, \mathbf{s}_j) \in \mathcal{E}\}$.

3. State transition function is the other end of th edge, $\mathbf{s}_{t+1} = f(\mathbf{s}_t, \mathbf{u}_t) = \text{end}(\mathbf{u}_t)$, where $\mathbf{s}_t = \text{start}(\mathbf{u}_t)$.

2.0.3 Representations of Graphs

(Chapter 23 of Introduction to Algorithms by Carmen et al)

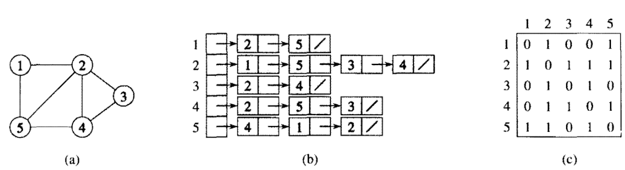


Figure 23.1 Two representations of an undirected graph. **(a)** An undirected graph G having five vertices and seven edges. **(b)** An adjacency-list representation of G. **(c)** The adjacency-matrix representation of G.

Undirected graph

```
# Programmatically you can represent a adjacency list as python lists
# Python lists are not linked lists, they are arrays under the hood.
G_adjacency_list = {
    1 : [2, 5], # node : list of neighbors
    2 : [1, 5, 3, 4],
    3 : [2, 4],
    4 : [2, 5, 3],
    5 : [4, 1, 2]
}

# Prefer to represent a matrix in python either as a list of lists or a numpy array
import numpy as np
G_adjacency_matrix = np.array([
    [0, 1, 0, 0, 1],
    [1, 0, 1, 1, 1],
    [0, 1, 0, 1, 0],
    [0, 1, 1, 0, 1],
    [1, 1, 0, 1, 0]
])

# Edge list is another possible representation
G_edge_list = [
    (1, 2), (1, 5),
    (2, 1), (2, 5), (2, 3), (2, 4),
    (3, 2), (3, 4),
    (4, 2), (4, 5), (4, 3),
    (5, 4), (5, 1), (5, 2)
]
```

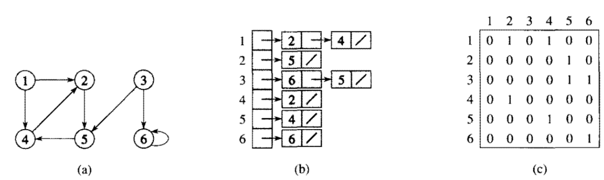


Figure 23.2 Two representations of a directed graph. **(a)** A directed graph G having six vertices and eight edges. **(b)** An adjacency-list representation of G. **(c)** The adjacency-matrix representation of G.

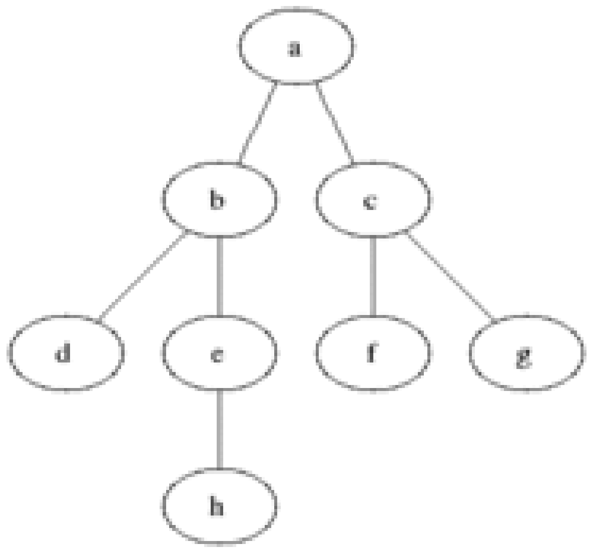
Directed graph representation

```
# Programmatically you can represent a adjacency list as python lists
# Python lists are not linked lists, they are arrays under the hood.
G_adjacency_list = {
    1 : [2, 4],
    2 : [5],
    3 : [6, 5],
    4 : [2],
    5 : [4],
```

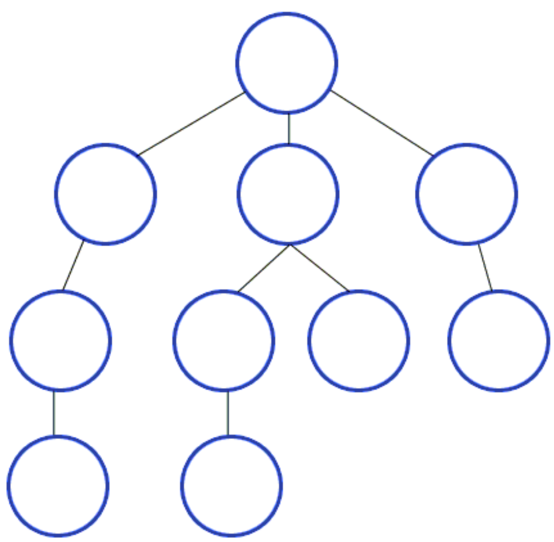
```
    5 : [6]
}

# Prefer to represent a matrix in python either as a list of lists or a numpy array
import numpy as np
G_adjacency_matrix = np.array([
    [0, 1, 0, 1, 0, 0],
    [0, 0, 0, 0, 1, 0],
    [0, 0, 0, 0, 1, 1],
    [0, 1, 0, 0, 0, 0],
    [0, 0, 0, 1, 0, 0],
    [0, 0, 0, 0, 0, 1]
])

# Edge list is another possible representation
G_edge_list = [
    (1, 2), (1, 4),
    (2, 5),
    (3, 6), (3, 5),
    (4, 2),
    (5, 6)
]

# Exercise 1

# Write a function that converts a graph in adjacency list format to adjacency matrix and vi
def adjacency_list_to_matrix(G_adj_list):
    G_adj_mat = None # TODO: Write code to convert to adj_mat
    return G_adj_mat

def adjacency_matrix_to_list(G_adj_mat):
    G_adj_list = None # TODO: Write code to convert to adj_mat
    return G_adj_list

# Use the above graphs to test
print(adjacency_list_to_matrix(G_adjacency_list))
print(adjacency_matrix_to_list(G_adjacency_matrix))
```

```
None
None
```

2.0.4 Graph Search algorithms



1. Breadth First Search
2. Depth First Search



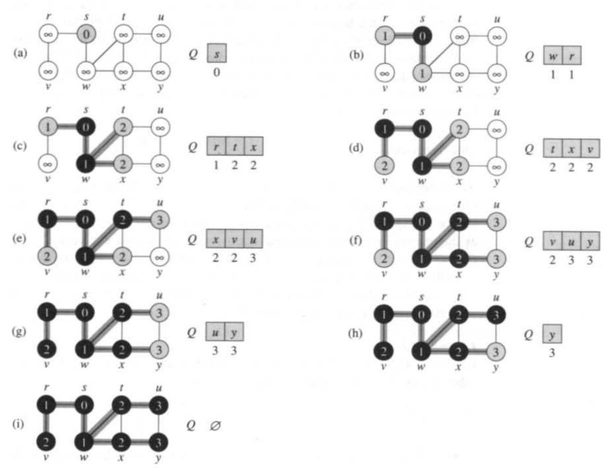


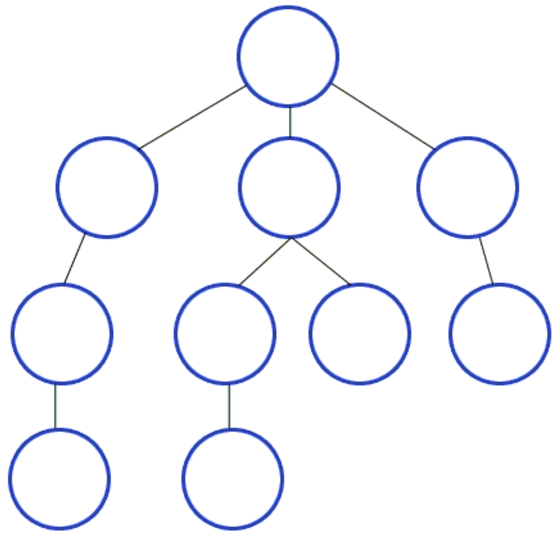
Figure 23.3 The operation of BFS on an undirected graph. Tree edges are shown shaded as they are produced by BFS. Within each vertex u is shown $d[u]$. The queue Q is shown at the beginning of each iteration of the **while** loop of lines 9–18. Vertex distances are shown next to vertices in the queue.

Breadth first search (BFS)

```
from queue import Queue, LifoQueue, PriorityQueue

graph = {
  's' : ['w', 'r'],
  'r' : ['v'],
  'w' : ['t', 'x'],
  'x' : ['y' ],
  't' : ['u'],
  'u' : ['y']
}

def bfs(graph, start, debug=False):
    seen = set() # Set for seen nodes (contains both frontier and dead states)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = Queue() # Frontier of unvisited nodes as FIFO
    node2dist = {start : 0} # Keep track of distances
    search_order = []
    seen.add(start)
    frontier.put(start)


    i = 0 # step number
    while not frontier.empty():          # Creating loop to visit each node
        if debug: print("%d) Q = " % i, list(frontier.queue), end='; ')
        if debug: print("dists = " , [node2dist[n] for n in frontier.queue])
        m = frontier.get() # Get the oldest addition to frontier
        search_order.append(m)
```

```
        for neighbor in graph.get(m, []):
            if neighbor not in seen:
                seen.add(neighbor)
                frontier.put(neighbor)
                node2dist[neighbor] = node2dist[m] + 1
            else:
                assert node2dist[neighbor] <= node2dist[m] + 1, 'this should not happen in B
                node2dist[neighbor] = min(node2dist[neighbor], node2dist[m] + 1)

        i += 1
    if debug: print("%d) Q = " % i, list(frontier.queue))
    return search_order, node2dist

print("Following is the Breadth -First Search order")
print(bfs(graph, 's', debug=True))    # function calling
```

```
Following is the Breadth -First Search order
0) Q =  ['s']; dists =  [0]
1) Q =  ['w', 'r']; dists =  [1, 1]
2) Q =  ['r', 't', 'x']; dists =  [1, 2, 2]
3) Q =  ['t', 'x', 'v']; dists =  [2, 2, 2]
4) Q =  ['x', 'v', 'u']; dists =  [2, 2, 3]
5) Q =  ['v', 'u', 'y']; dists =  [2, 3, 3]
6) Q =  ['u', 'y']; dists =  [3, 3]
7) Q =  ['y']; dists =  [3]
8) Q =  []
(['s', 'w', 'r', 't', 'x', 'v', 'u', 'y'], {'s': 0, 'w': 1, 'r': 1, 't': 2, 'x': 2, 'v': 2,
```



Depth first search

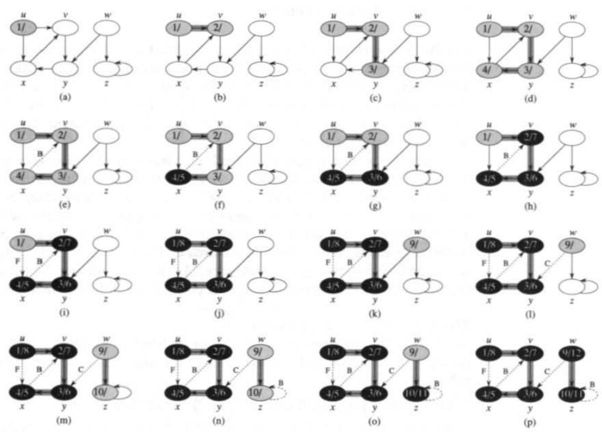


Figure 23.4 The progress of the depth-first-search algorithm DFS on a directed graph. As edges are explored by the algorithm, they are shown as either shaded (if they are tree edges) or dashed (otherwise). Nontree edges are labeled B, C, or F according to whether they are back, cross, or forward edges. Vertices are timestamped by discovery time/finishing time.

```
graph = {
  's' : ['w', 'r'],
  'r' : ['v'],
  'w' : ['t', 'x'],
  'x' : ['y' ],
  't' : ['u'],
  'u' : ['y']
```

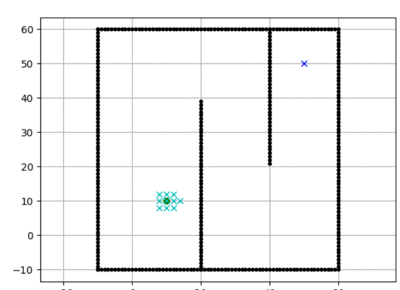
```
}

def dfs(graph, start, debug=False):
    seen = set([start]) # List for seen nodes (contains both frontier and dead states)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = LifoQueue() # Frontier of unvisited nodes as FIFO
    node2dist = {start : 0} # Keep track of distances
    search_order = [] # Keep track of search order
    frontier.put(start)


    i = 0 # step number
    while not frontier.empty():          # Creating loop to visit each node
        if debug: print("%d) Q = " % i, list(frontier.queue), end='; ')
        if debug: print("dists = " , [node2dist[n] for n in frontier.queue])
        m = frontier.get() # Get the oldest addition to frontier
        search_order.append(m)


        for neighbor in graph.get(m, []):
            if neighbor not in seen:
                seen.add(neighbor)
                frontier.put(neighbor)
                node2dist[neighbor] = node2dist[m] + 1
            else:
                node2dist[neighbor] = min(node2dist[neighbor], node2dist[m] + 1)
        i += 1
    if debug: print("%d) Q = " % i, list(frontier.queue))
    return search_order, node2dist

# Driver Code
print("Following is the Depth -First Search path")
print(dfs(graph, 's', debug=True))    # function calling

Following is the Depth -First Search path
0) Q =  ['s']; dists =  [0]
1) Q =  ['w', 'r']; dists =  [1, 1]
2) Q =  ['w', 'v']; dists =  [1, 2]
3) Q =  ['w']; dists =  [1]
4) Q =  ['t', 'x']; dists =  [2, 2]
5) Q =  ['t', 'y']; dists =  [2, 3]
6) Q =  ['t']; dists =  [2]
7) Q =  ['u']; dists =  [3]
8) Q =  []
(['s', 'r', 'v', 'w', 'x', 'y', 't', 'u'], {'s': 0, 'w': 1, 'r': 1, 'v': 2, 't': 2, 'x': 2,
```

Depth first search using recursion It is much easier to implement depth first search using recursion.

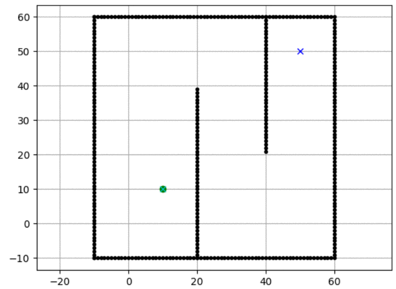
```
def dfs_recursive(graph, start,
                  seen=None,  # List for seen nodes (contains both frontier and dead states)
                  search_order=None, # Keep track of search order
                  node2dist=None, # Keep track of distances
                  debug=False):
    if seen is None: seen = set()
    if search_order is None: search_order = []
    if node2dist is None: node2dist = {start: 0}

    seen.add(start) # List for seen nodes (contains both frontier and dead states)
    search_order.append(start)

    for neighbor in graph.get(start, []):
        edge_dist = 1 # for a weighted graph, you might want to get edge_weight from graph
        if neighbor not in seen:
            node2dist[neighbor] = node2dist[start] + edge_dist
            search_order, node2dist = dfs_recursive(
                graph, neighbor, seen=seen,
                search_order=search_order,
                debug=debug, node2dist=node2dist)
        else:
            node2dist[neighbor] = min(node2dist[neighbor], node2dist[start] + edge_dist)
    return search_order, node2dist

# Driver Code
print("Following is the Depth -First Search path")
print(dfs_recursive(graph, 's', debug=True))    # function calling
```

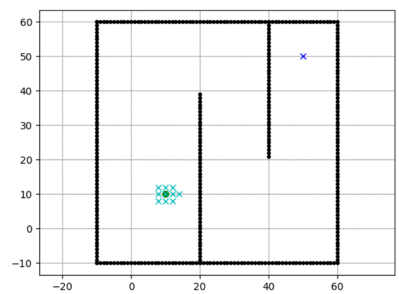
```
Following is the Depth -First Search path
(['s', 'w', 't', 'u', 'y', 'x', 'r', 'v'], {'s': 0, 'w': 1, 't': 2, 'u': 3, 'y': 3, 'x': 2,
```

2.0.5 Search order in BFS vs DFS vs Dijkstra

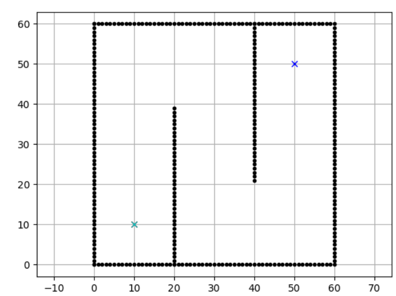


Breadth first search vs Depth first search





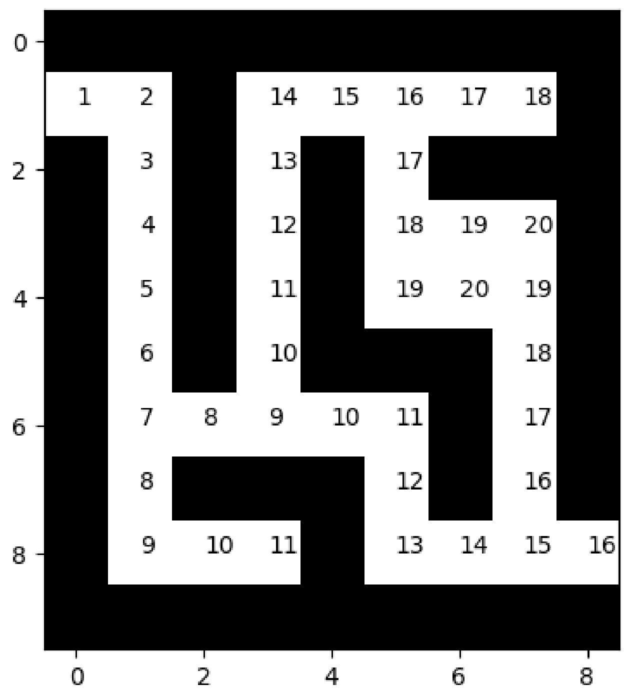
Breadth first search vs Dijkstra



2.0.6 Converting a maze search to a graph search

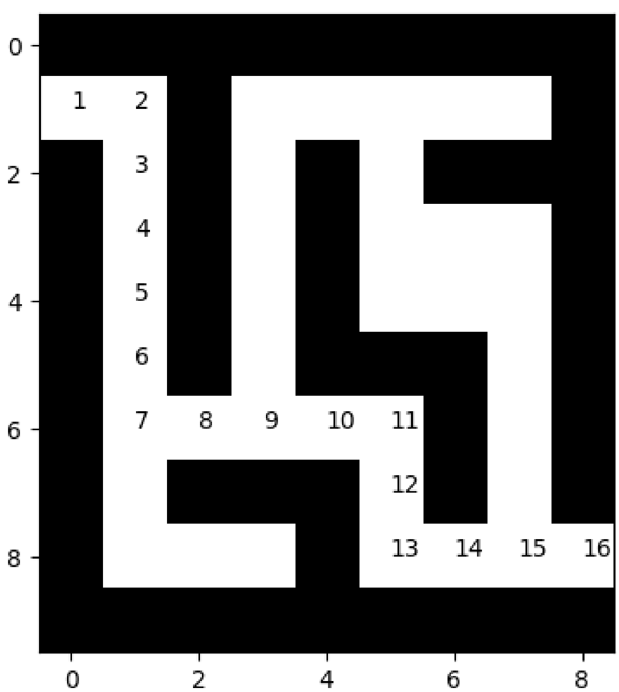
```
import matplotlib.pyplot as plt
import numpy as np
maze_str = \
"""
+++++++++
  +     +
+ + + +++
+ + +   +
+ + +   +
```

```
+ + +++ +
+     + +
+ +++ + +
+   +
+++++++++
"""

class Maze:
    def __init__(self, maze_str, freepath=' '):
        self.maze_lines = [l for l in maze_str.split("\n")
                           if len(l)]
        self.FREEPATH = freepath
        self.fig = None

    def get(self, node, default):
        (r, c) = node
        m_row = self.maze_lines[r]
        nbrs = []
        if c -1 >= 0 and m_row[c -1] == self.FREEPATH:
            nbrs.append((r, c -1))
        if c+1 < len(m_row) and m_row[c+1] == self.FREEPATH:
            nbrs.append((r, c+1))
        if r -1 >= 0 and self.maze_lines[r -1][c] == self.FREEPATH:
            nbrs.append((r -1, c))
        if r+1 < len(self.maze_lines) and self.maze_lines[r+1][c] == self.FREEPATH:
            nbrs.append((r+1, c))
        return nbrs if len(nbrs) else default
    init_plots = init_plots
    plot_maze = plot_maze
    plot_step = plot_step
    plot_path = plot_path
    animate_search_path = animate_search_path

import matplotlib.pyplot as plt
import matplotlib.animation as animation
import matplotlib as mpl
%matplotlib inline
mpl.rc('animation', html='jshtml')


maze = Maze(maze_str)
search_path, node2dist = bfs(maze, (1, 0)) # prints the order of search all the searched nod
maze.plot_maze()

maze.animate_search_path(search_path, node2dist)
```




```
def bfs_path(graph, start, goal):
    """
    Returns success and node2parent

    success: True if goal is found otherwise False
    node2parent: A dictionary that contains the nearest parent for node
    """
    seen = [start] # List for seen nodes.
    # Frontier is the boundary between seen and unseen
    frontier = Queue() # Frontier of unvisited nodes as FIFO
    node2parent = dict() # Keep track of nearest parent for each node (requires node to be h
    frontier.put(start)

    while not frontier.empty():          # Creating loop to visit each node
        m = frontier.get() # Get the oldest addition to frontier
        if m == goal:
            return True, node2parent

        for neighbor in graph.get(m, []):
            if neighbor not in seen:
                seen.append(neighbor)
                frontier.put(neighbor)
                node2parent[neighbor] = m
```

```
    return False, []

def backtrace_path(node2parent, start, goal):
    c = goal
    r_path = [c]
    parent = node2parent.get(c, None)
    while parent != start:
        r_path.append(parent)
        c = parent
        parent = node2parent.get(c, None) # Keep getting the parent until you reach the star
        #print(parent)
    r_path.append(start)
    return reversed(r_path) # Reverses the path

maze = Maze(maze_str)
start = (1, 0)
goal = (8, 8)
success, node2parent = bfs_path(maze, (1, 0), (8, 8))
path = backtrace_path(node2parent, (1, 0), (8, 8))
#print(list(path))
maze.plot_path(path) # Draws all the searched nodes
plt.show()
#node2parent
```



2.1 Dijkstra algorithm

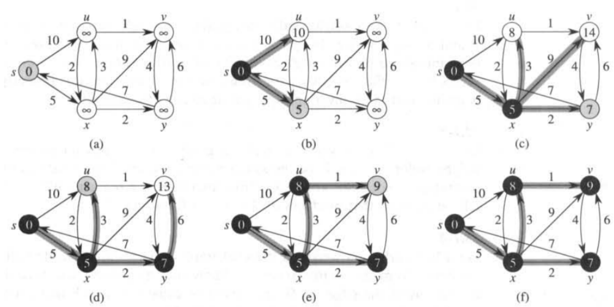


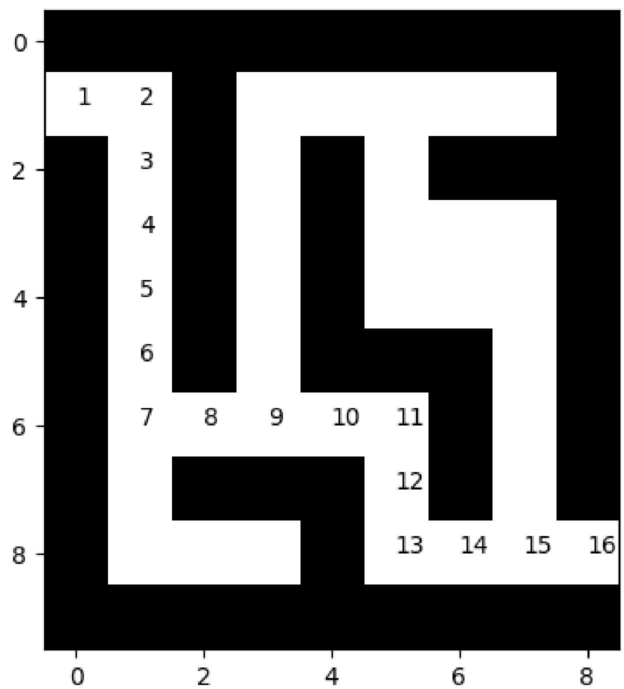
Figure 25.5 The execution of Dijkstra's algorithm. The source is the leftmost vertex. The shortest-path estimates are shown within the vertices, and shaded edges indicate predecessor values: if edge (u, v) is shaded, then $\pi[v] = u$. Black vertices are in the set S, and white vertices are in the priority queue $Q = V - S$. **(a)** The situation just before the first iteration of the **while** loop of lines 4–8. The shaded vertex has the minimum d value and is chosen as vertex u in line 5. **(b)–(f)** The situation after each successive iteration of the **while** loop. The shaded vertex in each part is chosen as vertex u in line 5 of the next iteration. The d and π values shown in part (f) are the final values.

2.1.1 PriorityQueue

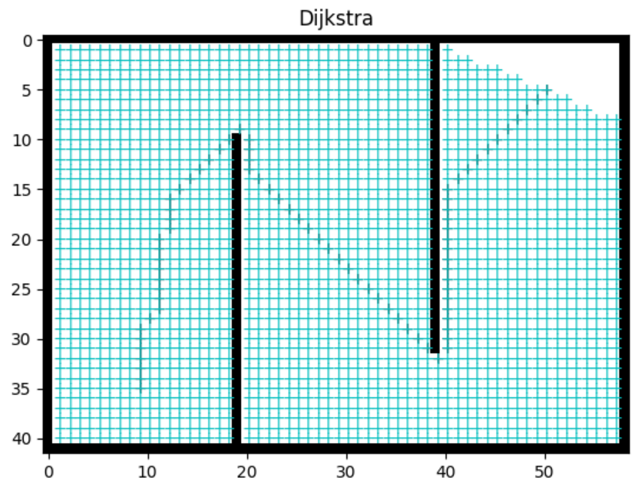
PriorityQueue returns the smallest (or the largest) item in the queue faster than other data structures

```
import itertools

class MazeD(Maze):
    def get(self, node, default):
        nbrs = Maze.get(self, node, default)
        return zip(nbrs, itertools.repeat(1))

maze = MazeD(maze_str)
success, search_path, node2parent, node2dist = dijkstra(maze, (1, 0), (8, 8))
print(success, node2parent)
if success:
    path = backtrace_path(node2parent, (1, 0), (8, 8))
    maze.plot_path(path) # Draws all the searched nodes
```

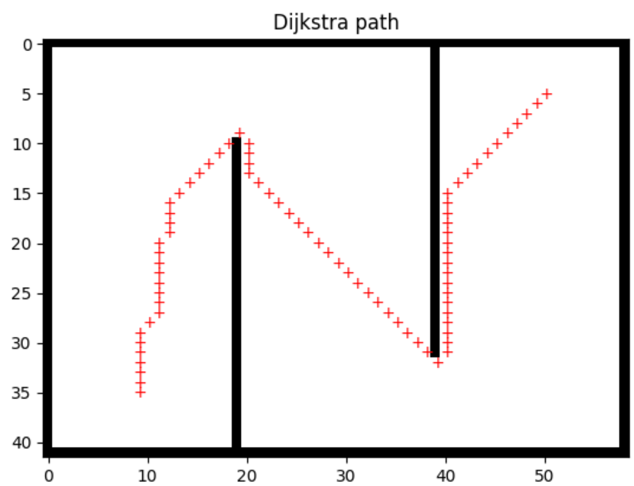
```
True {(1, 0): None, (1, 1): (1, 0), (2, 1): (1, 1), (3, 1): (2, 1), (4, 1): (3, 1), (5, 1):
```




```
maze = Maze8(maze_str)
success, search_path, node2parent, node2dist = dijkstra(maze, start_pos, goal_pos)
#print(success, search_path)
assert success
anim = maze.animate(search_path, title='Dijkstra')
path = backtrace_path(node2parent, start_pos, goal_pos)
#maze.init_plots(reinit=True)
path_plot = maze.plot_path(path, color='k') # Draws the traced shortest path
anim
```




```
path = backtrace_path(node2parent, (35, 9), (5, 50))
maze.init_plots(reinit=True, title='Dijkstra path')
maze.plot_path(path, color='r') # Draws the traced shortest path
plt.show()
```

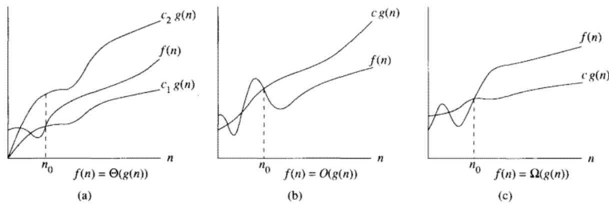


2.2 Computational complexity notation

Asymptotically tight bound $\Theta(g(n)) = \{f(n) :$ there exist positive constants $c_1, c_2,$ and n_o such that $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0\}$

Asymptotic upper bound $O(g(n)) = \{f(n) :$ there exist positive constants $c_2,$ and n_o such that $0 \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0\}$

Asymptotic lower bound $\Sigma(g(n)) = \{f(n) :$ there exist positive constants $c_1,$ and n_o such that $0 \leq c_1 g(n) \leq f(n)$ for all $n \geq n_0\}$



2.2.1 Computational complexity of BFS

```
# Write down the computational complexity of each line in big -O notation O()
# Assume the graph has |V| nodes and |E| edges
def bfs_barebones(graph, start):
    seen = {start} # Set for seen nodes (contains both frontier and dead states) # O(1)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = Queue() # Frontier of unvisited nodes as FIFO # O(1)
    frontier.put(start) # O(1)

    while not frontier.empty(): # Creating loop to visit each node # O(|V|)
        m = frontier.get() # Get the oldest addition to frontier # O(|V| * 1)

        for neighbor in graph.get(m, []): # O(|V| * |E|/|V|) = O(|E|)
            if neighbor not in seen: # O(|E| * 1)
                seen.add(neighbor) # O(|E| * 1)
                frontier.put(neighbor) # O(|E| * 1)
# The computational complexity of BFS is O(|V| + |E|)
```

2.2.2 Computational complexity of Dijkstra

```
# Write down the computational complexity of each line in big -O notation O()
# Assume the graph has |V| nodes and |E| edges
def dijkstra_barebones(graph, start):
    seen = {start} # Set for seen nodes (contains both frontier and dead states) # O(1)
    # Frontier is the boundary between seen and unseen (Also called the alive states)
    frontier = PriorityQueue() # Frontier of unvisited nodes as PriorityQueue # O(1)
    frontier.put(PItem(0, start)) # O(1)
    node2dist = {start: 0} # Keep track of cost to arrive at each node # O(1)

    while not frontier.empty(): # Creating loop to visit each node
        dist_and_node = frontier.get() # Get the smallest dist node # O(|V| * 1)
        m_dist = dist_and_node.dist
        m = dist_and_node.node

        for neighbor, edge_dist in graph.get(m, []): # O(|V| * |E|/|V|) = O(|E|)
```

```
            if neighbor not in seen: # O(|E| * 1)
                seen.add(neighbor) # O(|E| * 1)
                frontier.put(neighbor) # # O(|E| * log(|V|)) # for binary heap
                node2dist[neighbor] = m_dist + edge_dist # O(1)
            elif node2dist[neighbor] > m_dist + edge_dist: # O(1)
                node2dist[neighbor] = m_dist + edge_dist # O(1)

#  With Binary Heap based Prirority queue the complexity is O(|E|log |V| + |V|)

# The computational complexity of Dijkstra is O(|V|log(|V|) + |E|) when implemented
# using a Fibonacci heap based PriorityQueue
```

2.3 PriorityQueue (Heaps Chapter 7 of Carmen's intro to algorithms)

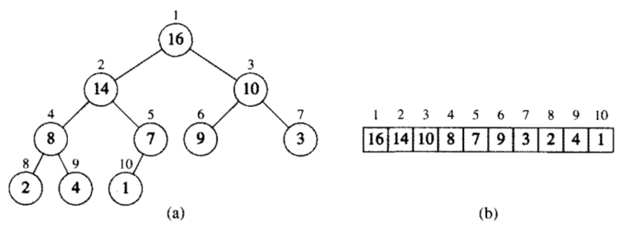


Figure 7.1 A heap viewed as **(a)** a binary tree and **(b)** an array. The number within the circle at each node in the tree is the value stored at that node. The number next to a node is the corresponding index in the array.

Heap property

1. $H[\text{Parent(i)}] \geq H[i]$
2. Parent(i) = ceil(i/2)
3. LeftChild(i) = 2i
4. RightRight(i) = 2i+1

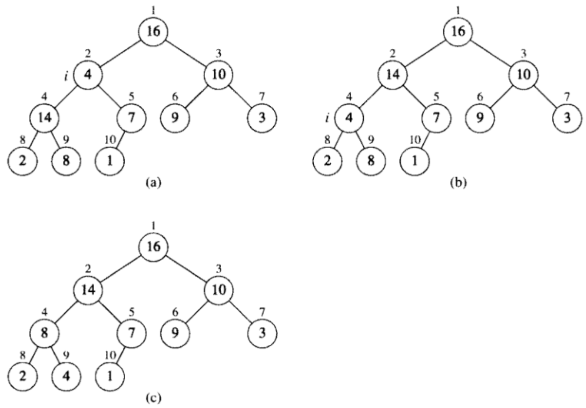


Figure 7.2 The action of HEAPIFY$(A, 2)$, where $heap\text{-}size[A] = 10$. **(a)** The initial configuration of the heap, with $A[2]$ at node $i = 2$ violating the heap property since it is not larger than both children. The heap property is restored for node 2 in **(b)** by exchanging $A[2]$ with $A[4]$, which destroys the heap property for node 4. The recursive call HEAPIFY$(A, 4)$ now sets $i = 4$. After swapping $A[4]$ with $A[9]$, as shown in **(c)**, node 4 is fixed up, and the recursive call HEAPIFY$(A, 9)$ yields no further change to the data structure.

Heapify

```
HEAPIFY(A, i)
l ← LEFT(i)
r ← RIGHT(i)
if l ≤ heap-size[A] and A[l] > A[i]
   then largest ← l
   else largest ← i
if r ≤ heap-size[A] and A[r] > A[largest]
   then largest ← r
if largest ≠ i
   then exchange A[i] ↔ A[largest]
        HEAPIFY(A, largest)
```

Heapify pseudocode

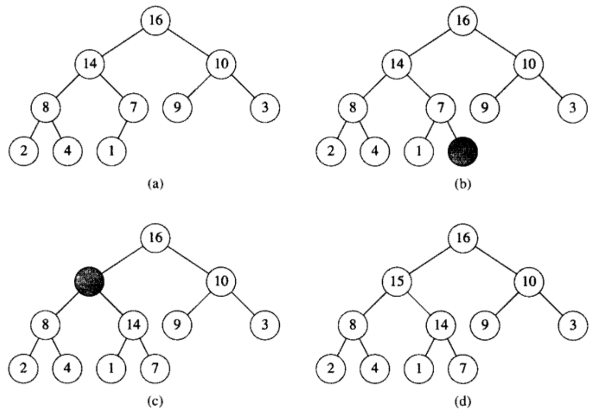


Figure 7.5 The operation of HEAP-INSERT. **(a)** The heap of Figure 7.4(a) before we insert a node with key 15. **(b)** A new leaf is added to the tree. **(c)** Values on the path from the new leaf to the root are copied down until a place for the key 15 is found. **(d)** The key 15 is inserted.

Heap Insert

```
HEAP-EXTRACT-MAX(A)
if heap-size[A] < 1
    then error "heap underflow"
max ← A[1]
A[1] ← A[heap-size[A]]
heap-size[A] ← heap-size[A] − 1
HEAPIFY(A, 1)
return max
```

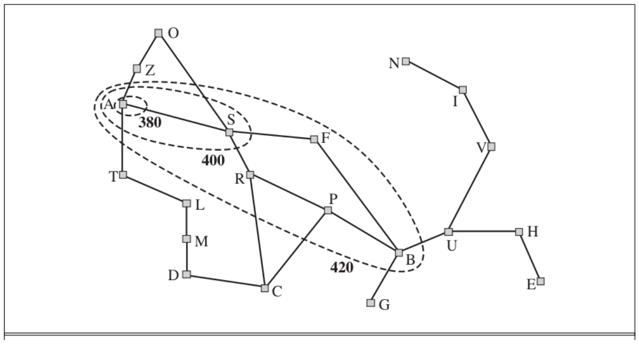
Heap Extract Max

Operation	find-min	delete-min	insert	decrease-key	meld
Binary[9]	$\Theta(1)$	$\Theta(\log n)$	$O(\log n)$	$O(\log n)$	$\Theta(n)$
Leftist	$\Theta(1)$	$\Theta(\log n)$	$\Theta(\log n)$	$O(\log n)$	$\Theta(\log n)$
Binomial[9][10]	$\Theta(1)$	$\Theta(\log n)$	$\Theta(1)$[a]	$\Theta(\log n)$	$O(\log n)$
Skew binomial[11]	$\Theta(1)$	$\Theta(\log n)$	$\Theta(1)$	$\Theta(\log n)$	$O(\log n)$[b]
Pairing[12]	$\Theta(1)$	$O(\log n)$[a]	$\Theta(1)$	$o(\log n)$[a][c]	$\Theta(1)$
Rank-pairing[15]	$\Theta(1)$	$O(\log n)$[a]	$\Theta(1)$	$\Theta(1)$[a]	$\Theta(1)$
Fibonacci[9][2]	$\Theta(1)$	$O(\log n)$[a]	$\Theta(1)$	$\Theta(1)$[a]	$\Theta(1)$
Strict Fibonacci[16]	$\Theta(1)$	$O(\log n)$	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
Brodal[17][d]	$\Theta(1)$	$O(\log n)$	$\Theta(1)$	$\Theta(1)$	$\Theta(1)$
2-3 heap[19]	$O(\log n)$	$O(\log n)$[a]	$O(\log n)$[a]	$\Theta(1)$	?

Heap runtimes

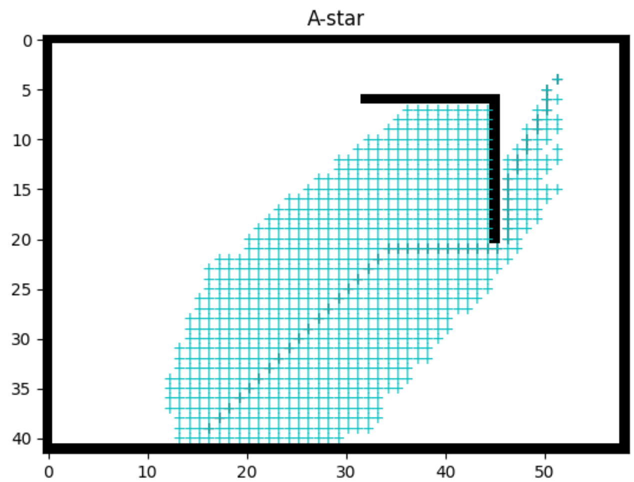
2.4 A-star (A*) algorithm

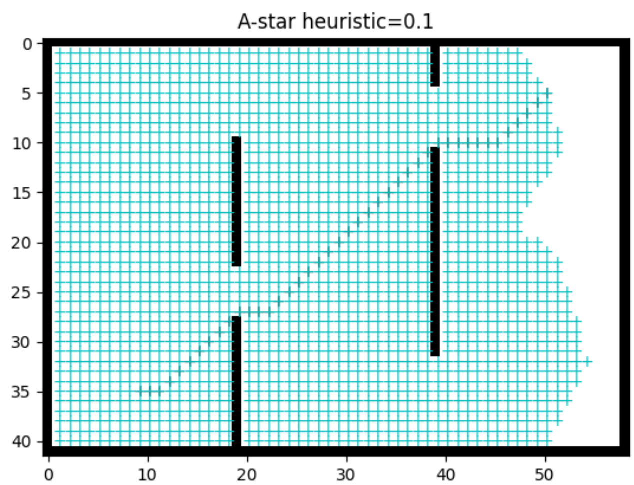
(Required reading: 3.5.2 of Russel and Norvig: Artificial Intelligence)

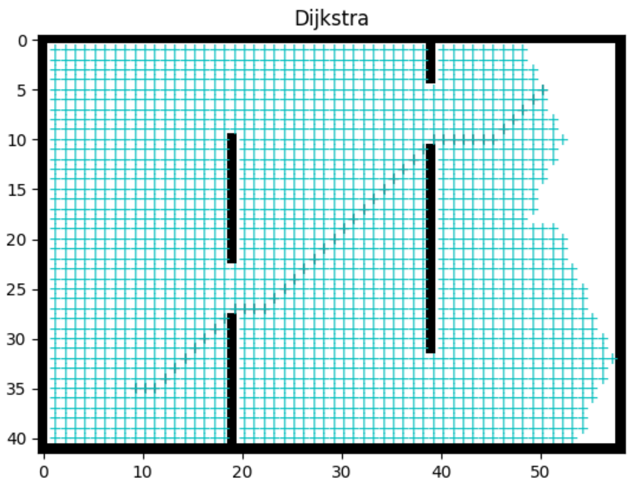



```
from hw2_solution import PriorityQueueUpdatable
import sys
def astar(graph, heuristic_dist_fn, start, goal, debug=False, debugf=sys.stdout):
    """
    edgecost: cost of traversing each edge

    Returns success and node2parent

    success: True if goal is found otherwise False
    node2parent: A dictionary that contains the nearest parent for node
    """
    seen = set([start]) # Set for seen nodes.
    # Frontier is the boundary between seen and unseen
    frontier = PriorityQueueUpdatable() # Frontier of unvisited nodes as a Priority Queue
    node2parent = {start : None} # Keep track of nearest parent for each node (requires node
    hfn = heuristic_dist_fn # make the name shorter
    node2dist = {start: 0  } # Keep track of cost to arrive at each node
    search_order = []
    frontier.put(PItem(0 + hfn(start, goal), start)) #   < - - - - - - - - - - - - - - Differe

    if debug: debugf.write("goal = "  + str(goal) + '\n')
    i = 0
    while not frontier.empty():          # Creating loop to visit each node
        dist_m = frontier.get() # Get the smallest addition to the frontier
        if debug: debugf.write("%d) Q = " % i + str(list(frontier.queue)) + '\n')
        if debug: debugf.write("%d) node = " % i + str(dist_m) + '\n')
        #if debug: print("dists = " , [node2dist[n.node] for n in frontier.queue])
        m = dist_m.node
        m_dist = node2dist[m]
        search_order.append(m)
        if goal is not None and m == goal:
            return True, search_order, node2parent, node2dist

        for neighbor, edge_cost in graph.get(m, []):
```

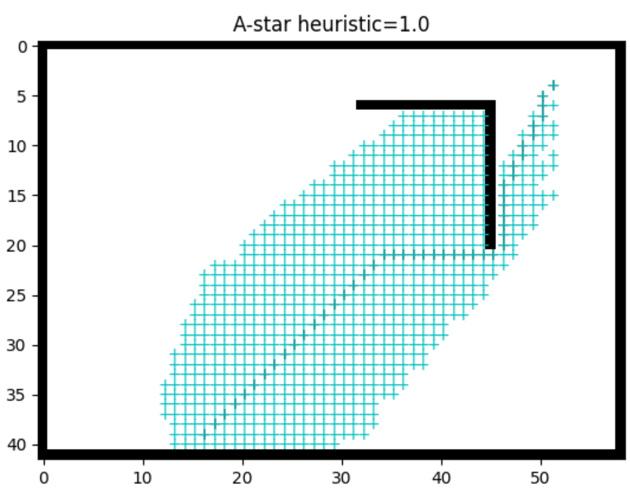
```
            old_dist = node2dist.get(neighbor, float("inf"))
            new_dist = edge_cost +  m_dist
            if neighbor not in seen:
                seen.add(neighbor)
                frontier.put(PItem(new_dist +  hfn(neighbor, goal), neighbor)) # < - - - - -
                node2parent[neighbor] = m
                node2dist[neighbor] = new_dist
            elif new_dist < old_dist:
                node2parent[neighbor] = m
                node2dist[neighbor] = new_dist
                # ideally you would update the dist of this item in the priority queue
                # as well. But python priority queue does not support fast updates
                # - - - - - - - - - - - - - Different from dijkstra - - - - - - - - - - - -
                old_item = PItem(old_dist + hfn(neighbor, goal), neighbor)
                if old_item in frontier:
                    frontier.replace(
                        old_item,
                        PItem(new_dist + hfn(neighbor, goal), neighbor))
        i += 1
    if goal is not None:
        return False, [], {}, node2dist
    else:
        return True, search_order, node2parent, node2dist

import math
from functools import partial

def euclidean_heurist_dist(node, goal, scale=1):
    x_n, y_n = node
    x_g, y_g = goal
    return scale*math.sqrt((x_n -x_g)**2 + (y_n - y_g)**2)
```


A-star
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50


A-star heuristic=0.1
0
5
10
15
20
25
30
35
40
0
10
20
30
40
50




```
maze = Maze8(maze_str)
success, search_path, node2parent, node2dist = astar(
    maze, partial(euclidean_heurist_dist, scale=1),
    start_pos, goal_pos)
#print(success, search_path)
assert success
anim = maze.animate(search_path, batch_size=20, title='A -star heuristic=1.0')
path = backtrace_path(node2parent, start_pos, goal_pos)
#maze.init_plots(reinit=True)
path_plot = maze.plot_path(path, color='k') # Draws the traced shortest path
anim
```



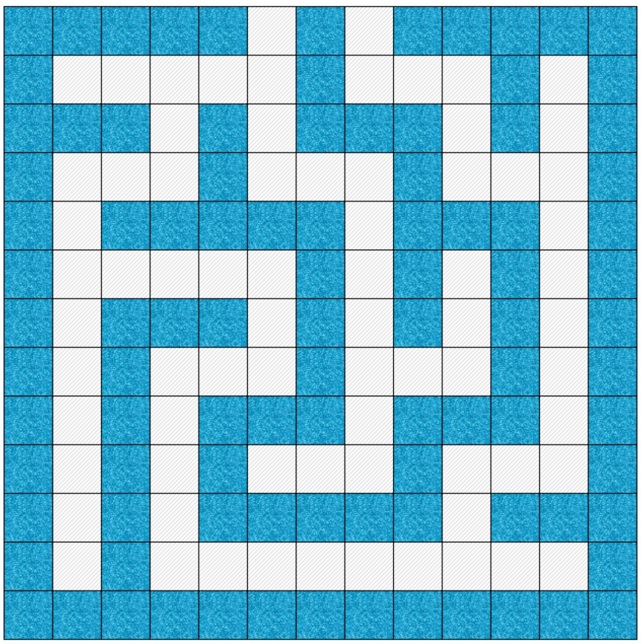
2.4.1 Admisibility and Consistency of heuristic function

1. An admissible heuristic is one that never overestimates the cost to reach the goal.

2. A heuristic h(n) is consistent if it satisfies the triangle inequality:

$h(\mathbf{s}_t) \leq c(\mathbf{s}_t, \mathbf{s}_{t+1}) + h(\mathbf{s}_{t+1})$.

2.5 Converting maze to grid to graph



2.6 Path Planning in continuous spaces

Step 1: Converting a continuous space into a graph
Step 2: Plan the path on the graph using A-star

2.6.1 Converting a continuous space into a graph

For convex polygon obstacles, you can use the corners as nodes. Add an edge only if the straight line segment connecting the nodes is completely outside the obstacles.

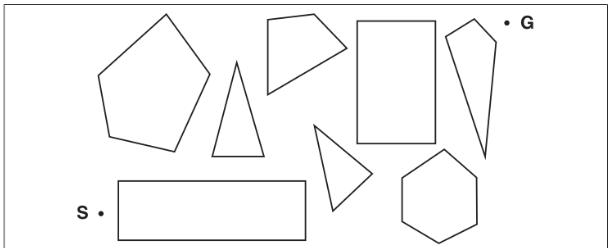
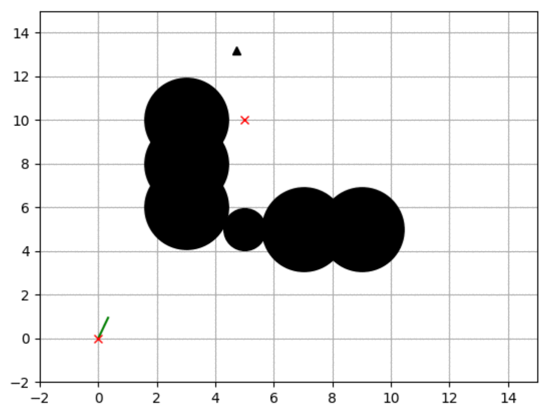


Figure 3.31 A scene with polygonal obstacles. S and G are the start and goal states.

2.6.2 Converting a continuous space into a graph

Sometimes you don't have polygonal obstacles, instead you only have a test for each point whether it is an obstacle or not. In such cases, you have to use a sampling based method like RRT, RRT-star, PRM (Probabilistic roadmaps) etc.



Rapidly exploring random trees (RRT)

Glossary

convex polygon A polygon such that any straight line segment connecting any two vertices lies entirely within the polygon.. 26